SMART EV
DASHBOARD PROTOTYPE DESIGN

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MEET JCORP MODEL SLE

Modern vehicle dashboards often present too much information at once, which can distract drivers and reduce road safety. As electric vehicles become more advanced, drivers need quick access to battery information, navigation, and driving assistance features without increasing cognitive workload.

Drivers frequently experience

- Information overload
- Difficulty locating important controls
- Reduced visibility during adversed weather
- diostractions while using navigation systems
- Anxiety regarding battery and charging status

OUR DESIGN PHILOSOPHY

Safety First

Prioritize critical driving information.

Minimal Distractions

Reduce unnecessary visual clutter.

Adaptive Interface

Interface responds to changing driving conditions.

Acessibility

High contrast and readable typography for all users.

USER RESEARCH & SYNTHESIS

AFFINITY DIAGRAM FOR THE CLUSTERS

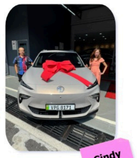
Cluster 1: Physical Controls & Ergonomic Flaws

Kenny John

Hard to navigate screens without physical buttons; accidental touches are a massive distraction.

Chloe

Adjusting the AC requires multiple taps through a screen menu,taking eyes off the road.


Cindy

Missed buttons because she expected physical ones but they were on the screen; experienced a learning curve to get used to it.

Jason

Missing physical buttons; capacitive touch causes accidental presses (hitting the radio instead of AC).

Axel Lau

Lacks physical buttons for frequently used creature comforts like HVAC (Aircond)

JT Lee

Dashboard screen is placed too far away from the driver's seat to operate comfortably.

Cluster 2: Cognitive Overload & Visual Clutter

Axel Lau

Feels overwhelmed by the sheer amount of information on some vehicle dashboards

Kenny John

Features like weather displays feel completely unnecessary and add to screen clutter.

Chloe

Overwhelmed by the combination of navigation details and real-time 3D vehicle visualization.

JT Lee

Sometimes cannot find functions in the apps because the interface isn't intuitive enough.

Jason

Distracted by unnecessary features like Twitter notifications popping up on the screen.

Cluster 3 : System & Software Performance

Cindy

Screen takes way too much time to boot up when car is still starting up

Cindy

Unresponsive and slow touchscreens forces her to touch the screen multiple times to use a single feature, causing distraction.

Melai

Small submenus making it difficult to press a button, as a result, majority of the time inaccuracy hit.

Cluster 4: Environmental Challenges

Axel Lau

Struggles with touch accuracy when it is raining, the road is slippery, and attention is split.

Jason

Severe sun glare from large, badly angled screens hits the driver's eyes.

Chloe

Pressing the wrong option mostly happens when driving on uneven, bumpy roads.

Kenny John

Missing the correct button on the touchscreen usually happens when it's raining.

Cluster 5: "Unsafe" Safety Features

JT Lee

Auto-braking system is too harsh, creating a fear of causing a rear-end accident with the car behind.

Kenny John

Lane Keep Assist uses too strong of a force to steer the car back; it feels scary.

Jason

"Nannified" safety features like Lane Keep Assist are annoying and have to be turned off every time the car starts.

Cluster 6: The "Ideal" Interface

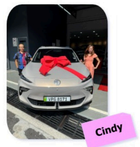
Kenny John

Ideal: A "no-frills" touchscreen. Get rid of unwanted features and only give users what they use everyday.

Chloe

Uses steering wheel shortcuts and voice commands to avoid the touchscreen entirely.

Ideal: Improve hands -free voice control.


Cindy

Avoids interacting with the screen entirely when it feels unsafe/unresponsive.

Ideal: Highly responsive, fast-booting screens.

Jason

Ideal: Keep common controls (AC, radio, indicators) physical, and fix screen angles to prevent glare.

JT Lee

Purposely drives further behind the car in front to prevent the auto-brake from triggering.

Ideal: A simpler, vital-stats display right in front of the driver.

Axel Lau

Ideal: Would only use AI and voice assistance if it reaches a highly matured, reliable stage.

AXEL LAU

Age	30
Driving Environment	Heavy city traffic and frequent highway travel for meetings. Often deals with bright sun glare
Occupation	Freelance Graphic Designer
Income Level	RM 8,000+

Background

A busy executive who views driving as a focused task. He relies on smartphone mirroring and is deeply frustrated by the trend of complex, laggy touchscreens replacing simple physical buttons.

Core Goals

- Adjust cabin comfort instantly without looking away from the road.
- Distraction-free environment prioritizing driving over "smart" apps.
- Zero-lag system boot-up and touch response.

System Usage Metrics

Physical Button Need

Touchscreen Reliance

Behavioral Workarounds

- Ignores native OS/voice tech; uses plugged-in phone only.
- Manually disables aggressive safety features before every drive.

Pain Points

- **System Lag**: Slow boot times and unresponsiveness cause dangerous distraction.
- **Lack of Tactility**: Misses flat screen buttons on bumpy roads.
- **Ergonomics & Clutter**: Sun glare on screens; annoyed by useless weather/social apps.
- **Intrusive Safety**: Lane Keep Assist is overly aggressive and jarring.

Ideal Solution: Lightning-fast OS, return of physical HVAC/volume dials, and a simplified dashboard.

Quote: "I just need the screen to boot up fast and let me change the aircon without looking away. I don't need a weather app going 90km/h."

CHLOE YAP

Age	38
Driving Environment	Daily city commuting, stressed by night driving and heavy rain
Occupation	Senior Accountant
Income Level	RM 6,500

Background

An EV driver who loves minimalist tech aesthetics but struggles with screen-only interfaces. She has actively adapted her driving habits to safely operate her car's deep menus and hyper-sensitive safety features.

Core Goals

- Monitor vital EV stats (range, speed, blindspots) at a quick glance.
- Safely control cabin environment during heavy rain/traffic.
- Maximize hands-free/tactile alternatives to the main screen

System Usage Metrics

Digital UI Adoption

In-Motion Screen Use

Behavioral Workarounds

- Relies exclusively on steering wheel buttons and voice commands while moving.
- Leaves massive gaps in traffic to prevent auto-brake from triggering.

Pain Points

- **Menu Depth**: Basic AC adjustments require 2-3 dangerous screen taps.
- **Visual Overload**: 3D vehicle UI is too complex to process quickly.
- **Overly Sensitive Tech**: Auto-braking engages too harshly, causing rear-end anxiety.
- **Touch Accuracy**: Struggles to hit correct digital buttons on uneven roads.

Ideal Solution: Flatter menu structures (fewer taps) and a secondary Heads-Up Display (HUD) for vital stats.

Quote: "I love the minimalist look, but tapping three times to turn up the fan in the rain is terrifying. I just use steering wheel buttons."

Cindy

Age	46
Driving Environment	Daily city commuting, stressed by night driving and heavy rain
Occupation	Corporate director
Income Level	RM 8,000 and above

Background

An EV driver who embraces new tech and values convenient, modern features like Apple CarPlay, but is frustrated by sluggish system performance, exceptionally long boot times and unresponsiveness that threatens her attention on the road.

Core Goals

- Seamlessly access modern convenience features and standard smartphone mirroring apps.
- Maintain focus on busy city roads without waiting for lagging system interfaces.
- Quickly adapt to screen-based button layouts despite initial learning curves

System Usage Metrics

Digital UI Adoption

Physical Button

Behavioral Workarounds

- Repeatedly multi-taps the display out of habit to force sluggish and unresponsive UI features to finally register her inputs.
- Reduces frequency with more inaccesible features while driving

Pain Points

- **Sluggish Boot Times**: System infotainment systems take far too long to turn on and boot up upon starting the vehicle.
- **Laggy Interface**: Unresponsive software forces multiple screen taps, creating a highly distracting and unsafe driving environment.
- **Hidden Physical Layouts**: Digital screen buttons replace layouts that logically feel like they should be physical, creating an unnecessary learning curve.

Ideal Solution: Build highly responsive, lag-free, and quick-booting infotainment hardware that executes commands instantly to protect driver attention.

Quote: "When using screens that are unresponsive, it would be very distracting especially when driving, and after a while it can feel unsafe, so I just don't interact."

JASON

Age	25
Driving Environment	Daily city commuting, stressed by night driving and heavy rain
Occupation	Journalist
Income Level	RM6,500

Background

A driver who heavily relies on smartphone interfaces for touch navigation but is deeply frustrated by poor screen ergonomics. He struggles with excessive sun glare from poorly angled displays, physical injuries from sharp screen edges, and overly sensitive safety features.

Core Goals

- Use familiar, touch-optimized smartphone interfaces via Android Auto or Apple CarPlay.
- Ensure primary touchpoints (AC, radio, shifters) remain entirely tactile.
- Prevent distracting notifications and screen glare from disrupting his view.

System Usage Metrics

Physical Button

In-Motion Screen Use

Behavioral Workarounds

- Leaves safety features active on shorter trips since turning them off every single time is too tedious.
- Simply puts up with touch inaccuracies and ergonomic annoyances while driving.

Pain Points

- **Screen Placement Flaws:** Large screens set at bad angles that causes blinding sun glare and sharp edges that physically hit his knee.
- **Capacitive Touch Errors:** Flat touch controls with no physical feel, causing accidental inputs while driving
- **Forced System Resets:** Annoying safety features (like Lane Keep Assist) aggressively force themselves back on every time the car starts.

- **Ideal Solution: Retain screens for mapping, but mandate physical controls for core common touchpoints (AC, radio, indicators, shifters); optimizing screen angles to entirely kill sun glare.**
- **Quote: "Keep controls physical; capacitive touch for normal functions is just an ergonomic flaw."**

KENNY JOHN

Age	30
Driving Environment	Daily city commuting, stressed by night driving and jammed road.
Occupation	Salesman
Income Level	RM 6,500

Background

Travels around for roadshow events and test drives. He usually does city driving and faces jams constantly when moving cars around or on his daily commute.

Core Goals

- Access mapping and music quickly without visual clutter.
- Utilize dedicated tactile layouts for everyday dashboard tasks.
- Maintain stable lane tracking without aggressive steering intervention.

System Usage Metrics

Physical Button

In-Motion Screen Use

Behavioral Workarounds

- Relies exclusively on steering wheel buttons and voice commands while moving.
- Leaves massive gaps in traffic to prevent auto-brake from triggering.

Pain Points

- **Zero Tactile Controls:** Screen-only setups cause accidental taps and massive distraction while driving.
- **Unnecessary Feature Bloat:** Useless dashboard widgets (like weather) clutter the view.
- **Intrusive Lane Keep Assist:** Safety tech steering pull feels overly strong and scary in action.

Ideal Solution: More physcial button for features that mostly require readjustment.

Quote: "I found the information is a bit too much. Sometimes it felt cluttered.

JT LEE

Age	25
Driving Environment	Daily city commuting, stressed by night driving and heavy rain
Occupation	Senior Accountant
Income Level	RM6,500

Background

An EV driver who values minimalist tech aesthetics but struggles with central touchscreen positioned too far from the driver's seat. While the clean UI prevents visual overload, he finds navigating non-intuitive app menus physically awkward and unsafe while in motion.

Core Goals

- Glanceable Monitoring: Track map steps and vital blindspot alerts instantly.
- Tactile Steering Sync: Maximize wheel-button use to avoid reaching for the screen.
- Defensive Buffer Space: Keep extra road gaps to counter sudden automatic braking.

System Usage Metrics

In-motion Screen Use

Digital UI Adoption

Behavioral Workarounds

- Relies exclusively on steering wheel buttons and voice commands while moving.
- Leaves massive gaps in traffic to prevent auto-brake from triggering.

Pain Points

- **Ergonomic Screen Distance:** Center display sits too far away to reach comfortably while moving.
- **Buried App Settings:** Non-intuitive UI layouts force heavy reliance on search bars.
- **Harsh Auto-Brake Sync:** Aggressive automatic braking triggers constant fear of rear-end crashes.

- **Ideal Solution: Flatter menu structures (fewer taps) and a secondary Heads-Up Display (HUD) for vital stats.**
- **Quote: "Tesla's design is minimalist so I don't feel overwhelmed, but the dashboard is far away, making it difficult to operate while driving."**

MELAI BELINGAU

Age	65
Driving Environment	Rural roads, mostly in the morning and under the sun.
Occupation	Pastor
Income Level	RM4,000

Background

A SUV owner who often went to travel across Sarawak. He struggle with the button sizing and found it hard to navigate especially when readjusting the air-conditioning.

Core Goals

- Easily adjust cabin temperature without struggling with tiny digital interface targets.
- Ensure screen remains clear and legible under harsh daytime sun glare.
- Ensure simplicity in usage entirely on navigating unpredictable rural roads.

System Usage Metrics

Digital UI Adoption

In-Motion Screen Use

Behavioral Workarounds

- Relies exclusively on steering wheel buttons and voice commands while moving.
- Leaves massive gaps in traffic to prevent auto-brake from triggering.

Pain Points

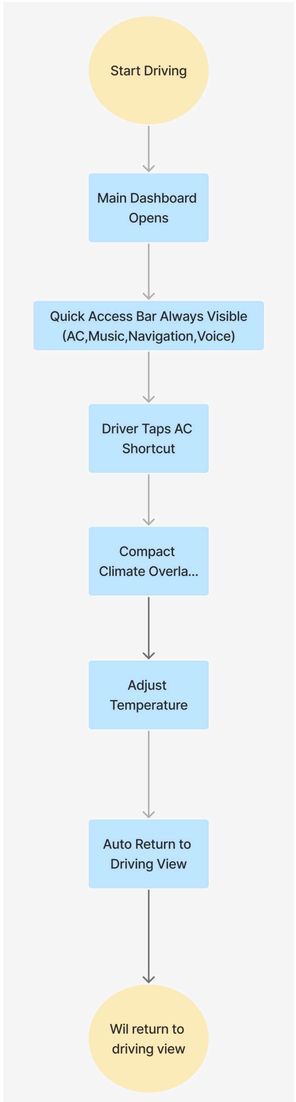
- **Small Button Sizing:** On-screen targets and digital icons are too small and tightly packed; leads to inaccuracy when press while moving.
- **Sun Glare Washout:** High daytime sun exposure on rural routes makes the glossy central screen difficult to read at a glance.
- **Complex Air-Con Submenus:** Adjusting basic climate controls forces him to dive into deep, non-intuitive interface layers.

Ideal Solution: Flatter menu structures (fewer taps) and a secondary Heads-Up Display (HUD) for vital stats.

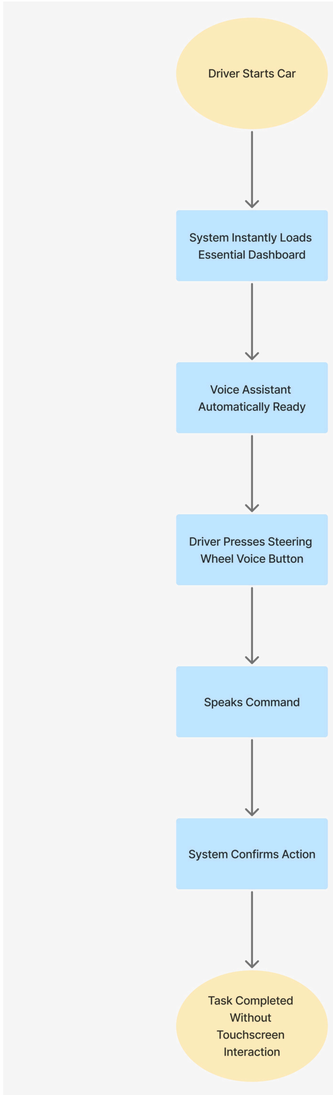
Quote: "I love the minimalist look, but tapping three times to turn up the fan in the rain is terrifying. I just use steering wheel buttons."

USER FLOW DIAGRAM

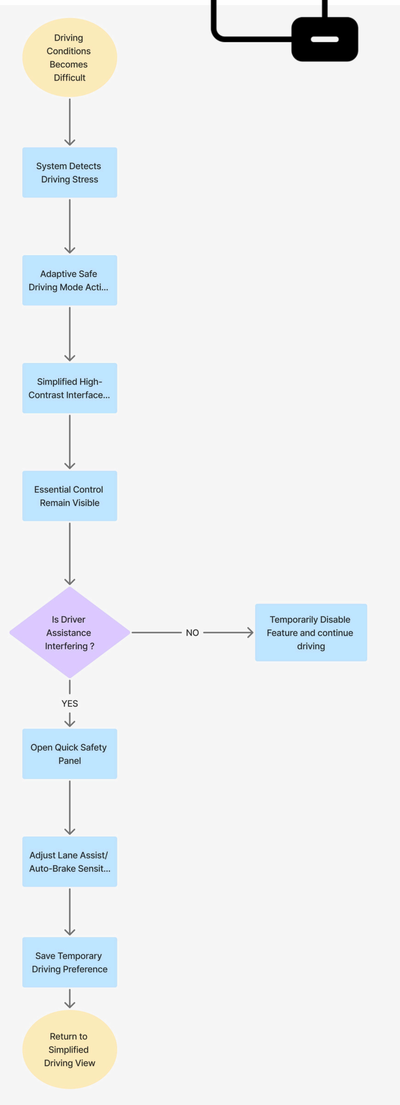
Adjust AC quickly when Driving



Perform Vehicle Actions Using Voice Commands



Manage safety features without Distractions



LOW-FI DESIGN

SCREEN 1: MAIN DASHBOARD

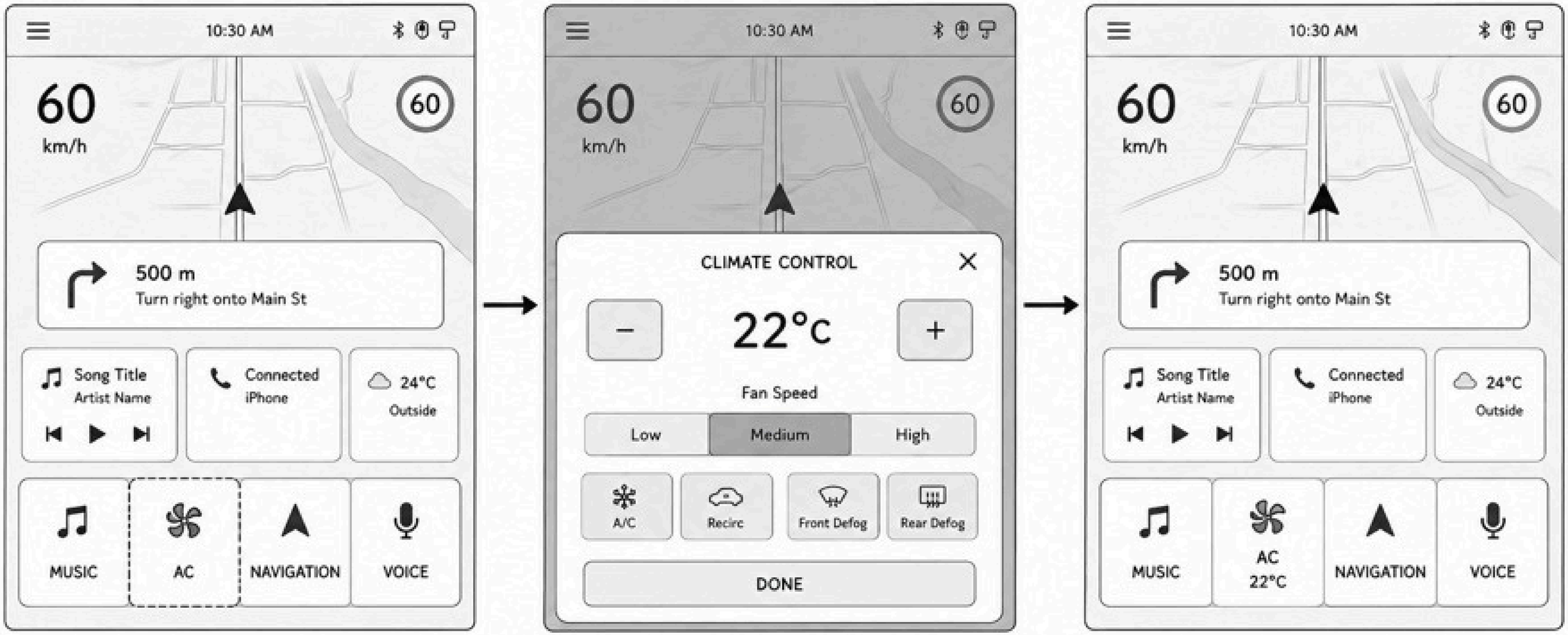
Simplified home screen with persistent quick access bar.

SCREEN 2: QUICK CONTROLS OVERLAY

Compact climate control overlay appears after tapping AC.

SCREEN 3: ADJUSTED STATE

Overlay closes and dashboard returns with subtle confirmation.



Adjust A/C / Media / Navigation quickly while driving

Perform vehicle actions using voice command

1. SCREEN 1: NORMAL DRIVING UI

Regular driving interface.

2. SCREEN 2: ADAPTIVE SAFE MODE ACTIVATED

System detects stress and switches to simplified, high-contrast view.

3. SCREEN 3: SAFETY SETTINGS OVERLAY

Quick panel to adjust or override assistance features.

Manage safety features without distractions

HIGH-FIDELITY PROTOTYPE

Sunday, 15 July 2026

JCORP MODEL SLE

2026 edition

2.912 km	5.000 km	82 km/h
Traveled	Maintenance	Speed Avg

Directions to Mons Hotel — 36 km — 15 km

In 140 m — Turn Right

Then

Voice Assistant

Air Conditioner

Temperature 23.5°C

100% Charged

348 km Left

P R N D **S**

I THINK

Tyler, The Creator

01:43 — 03:50

Home

Sunday, 15 July 2026

JCORP MODEL SLE

2026 edition

2.912 km	5.000 km	82 km/h
Traveled	Maintenance	Speed Avg

P R N D

S

100%
348 km

Voice Assistant

I THINK
Tyler, The Creator

Air Conditioner

A/C

Temperature
23.5°C

Directions to
Mons Hotel
36 km
15 km

In 140 m
Turn Right

Then

Home

Sunday, 15 July 2026

JCORP MODEL SLE

2026 edition

2.912 km	5.000 km	82 km/h
Traveled	Maintenance	Speed Avg

P R N D

S

100%
348 km

Voice Assistant

I THINK
Tyler, The Creator

Air Conditioner

A/C

Temperature
23.5°C

Directions to Mons Hotel
36 km
15 km

In 140 m
Turn Right

Then

Home

Sunday, 15 July 2026

Voice Assistant

Listening...

Home

Sunday, 15 July 2026

Voice Assistant

Action Completed

Home

Sunday, 15 July 2026

HEAVY RAIN DETECTED

Adaptive Interface Activated

[Interface Simplified]

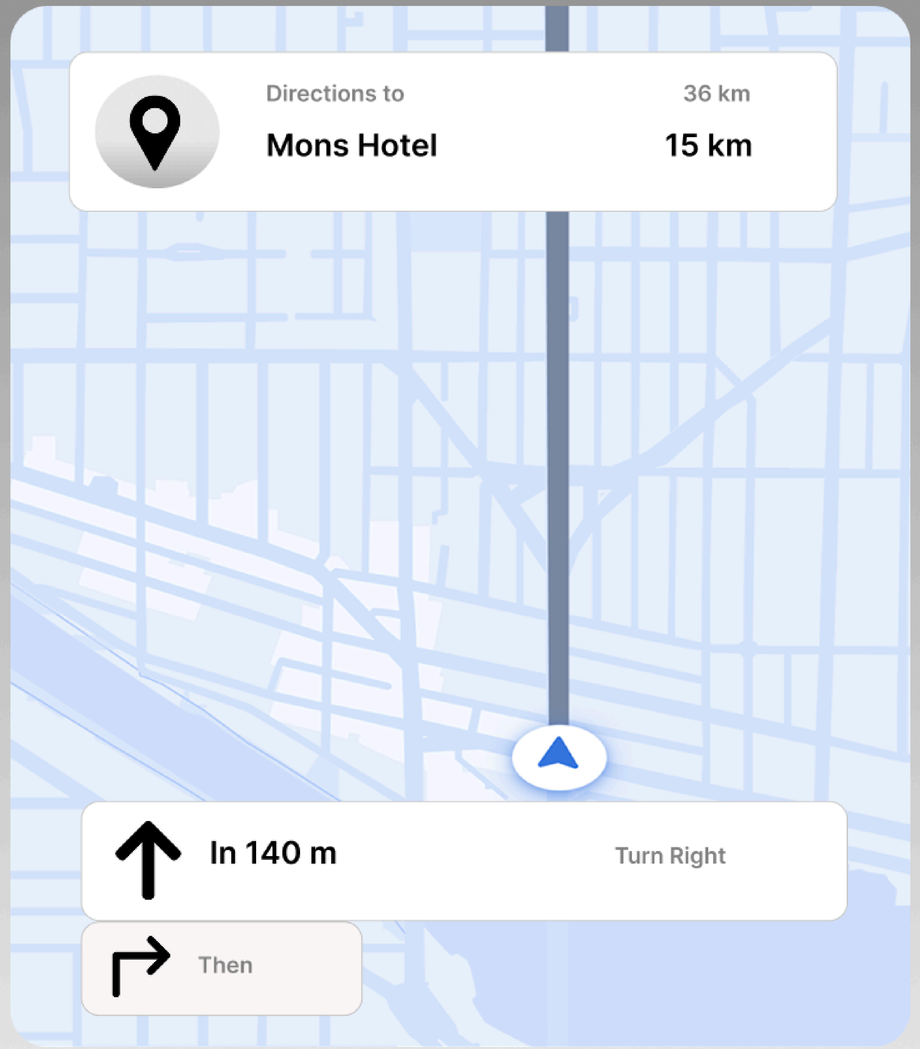
[Contrast Increased]

[Essential Controls Prioritized]

Home

Sunday, 15 July 2026

PRNDS



Voice Assistant

Lane Assist
ON

Safety Alerts
OFF

Auto Brake
ON

Safety Controls

EXIT

SAFETY CONTROLS

Lane Assist Sensitivity

Low —○———————— High

Auto Brake Sensitivity

Low ————————○—— High

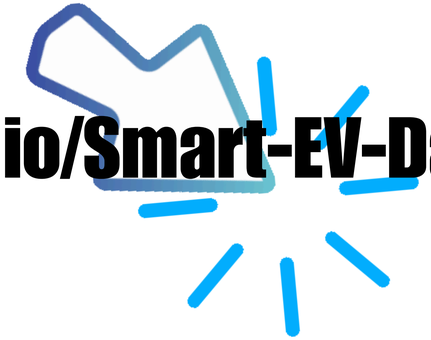
Driver Monitoring Alerts

TEMPORARY OVERRIDE (15 MIN)

EXIT

INTERACTIVE PROTOTYPE

https://zamarulnurddin.github.io/Smart-EV-Dashboard-Prototype/index.html



Link to Interactive prototype

USABILITY TESTING

1.Executive Summary

The purpose of this usability report is kind of to evaluate the user experience, interface clarity, and how fast tasks get finished, on the Smart EV Dashboard Prototype. The testing was done using a qualitative Think-Aloud method, but also with quantitative performance metrics, which is like measuring timing and accuracy. A total of 7 participants completed 6 structured task categories, an d these covered core vehicle status, media controls, climate tweaks, device connectivity, voice interaction, and also safety settings.

In general the results show that the core dashboard and entertainment parts feel very intuitive, but there is some significant user friction inside the advanced safety configurations and for the more precise climate adjustments. This report brings forward the key performance data and it also lays out actionable design recommendations, to optimize the dashboard so it supports safer, distraction-free driving.

2. Methodology & Demographics

Testing Method: Moderated Usability Testing utilizing the Think-Aloud protocol. Participants were instructed to vocalize their thoughts, expectations, and frustrations while completing tasks.

Metrics Tracked:
Quantitative: Task Success Rate (Yes/No), Time-on-Task (seconds), System Usability Scale (SUS), and Error Counts.
Qualitative: Observed Hesitation Points and open-ended participant feedback.
Participants: 6 Tester were chosen by various background.

Evaluation Tools: The live web prototype was hosted via GitHub Pages, task matrices were recorded manually via Table 1, and System Usability Scale (SUS) questions was captured using a structured Google Form .
For the google form,

there is two part which is **Part 1: Participant Demographics** where user basic information were taken such as Name ,age ,Driving Frequency ,Tech Comfort Level and experience using a touchscreen dashboard for an EV Car

While **Part 2: System Usability Scale (SUS) Statements** where Participants rated their agreement with the 10 standard System Usability Scale (SUS) psychometric items using a 5-point Likert scale ranging from **1 (Strongly Disagree)** to **5 (Strongly Agree):**

1.I think that I would like to use this system frequently.
2.I found the system unnecessarily complex.
3.I thought the system was easy to use.
4.I think that I would need the support of a technical person to be able to use this system.
5.I found the various functions in this system were well integrated.
6.I thought there was too much inconsistency in this system.
7.I would imagine that most people would learn to use this system very quickly.
8.I found the system very cumbersome to use.
9.I felt very confident using the system.
10.I needed to learn a lot of things before I could get going with this system.

3.Quantitative Performance Summary Matrix

3.1. Dashboard Usability and User Performance Metrics

Dashboard Area	Success Rate	Average Time	Hesitation Points	Error Count	Performance Translation
Core Dashboard & Vehicle Status	91.7%	40.18 seconds	1	0	Performed well overall and most users could locate battery range, distance travelled, and maintenance information. However, some users had difficulty identifying the current gear display, showing that secondary vehicle information should be made more visible.
Media & Music Control	100%	30.31 seconds	0	0	One of the strongest sections where users completed all music-related tasks successfully without hesitation or errors, showing that the media controls were intuitive and easy to understand.
Air Conditioner	93.3%	70.85 seconds	7	1	The highest time and hesitation count although most tasks were completed, users hesitated when opening the air-conditioner screen and locating the seat heater function. This shows that the climate control interface requires clearer entry points and button labels.

Connectivity & Devices	100%	29.02 seconds	0	0	This section performed strongly. Users were able to identify Bluetooth devices, connected devices, and signal strength without errors. This suggests that the connectivity layout was clear and straightforward.
Voice Assistant	100%	18.47 seconds	2	0	The voice assistant was easy to activate, but hesitation occurred because some users were unsure when the assistant started or stopped listening. This indicates that clearer voice feedback or status indicators are needed.
Settings & Secondary Controls	100%	50.97 seconds	0	0	Users successfully completed safety-related tasks, including driver monitoring alerts, auto brake sensitivity, and temporary override mode. The longer time suggests that this section required more navigation steps, but it did not cause major confusion.

3.2 System Usability Scale (SUS) Analysis

After think-aloud testing ended, the System Usability Scale (SUS) was given to all 7 participants. SUS is a validated 10-item questionnaire that ends up with one single score, from 0 to 100, and where higher scores mean better perceived usability (Brooke, 1996). The industry benchmark average is 68, which is basically the 50th percentile, across thousands of systems tested since 1986.

Calculation Methodology :

SUS scoring followed the standard protocol:

- Odd-numbered items (1,3,5,7,9): Contribution = (user response – 1)
- Even-numbered items (2,4,6,8,10): Contribution = (5 – user response)
- Final score: (Sum of 10 contributions) × 2.5

Individual Participant Scores

	Participant	SUS Score
1	MUHAMMAD DANISH BIN IDRIS	100.0
2	Mohamad syafiq bin Jamail	100.0
3	Davin Liew Jia Poh	72.5
4	Khairul nizan bin ramzul	72.5
5	Mugilan A/L Vasanthan	67.5
6	Ahmad Danish bin Samsudin	62.5
7	ELFAZER EZRA AGUH	60.0

Table 2: *Breaks down the System Usability Scale (SUS) scores obtained by each of the seven individual study participants, arranged from highest to lowest.*

Aggregate Results

Metric	Value
Number of participants (n)	7
Highest score	100.0
Lowest score	60.0
Mean SUS Score	**76.4**
Standard deviation	15.8
Industry benchmark	68.0

Table 4: *Summarises the descriptive metrics of the evaluation, highlighting the sample size (n = 7), data spread, and a comparison between the mean SUS score and the standard industry benchmark.*

Interpretation:

The obtained mean SUS score of 76.4 exceeds the industry benchmark of 68 by 8.4 points. According to standardized SUS grading criteria:

SUS Score Range	Letter Grade	Percentile Rank	Descriptive Usability Rating
85–100	A	90–100	Excellent
70–84	B	70–89	Good
50–69	C	30–69	Marginal
25–49	D	10–29	Poor
0–24	F	0–9	Unusable

Table 5: Maps out the standardized reference tiers for converting raw SUS scores into letter grades, percentile ranks, and descriptive usability ratings.

The EV dashboard achieved a **"Good" (Grade B) rating**, placing it above the 50th percentile but below the top 10% threshold of 80.3.

- **Key Observations:**

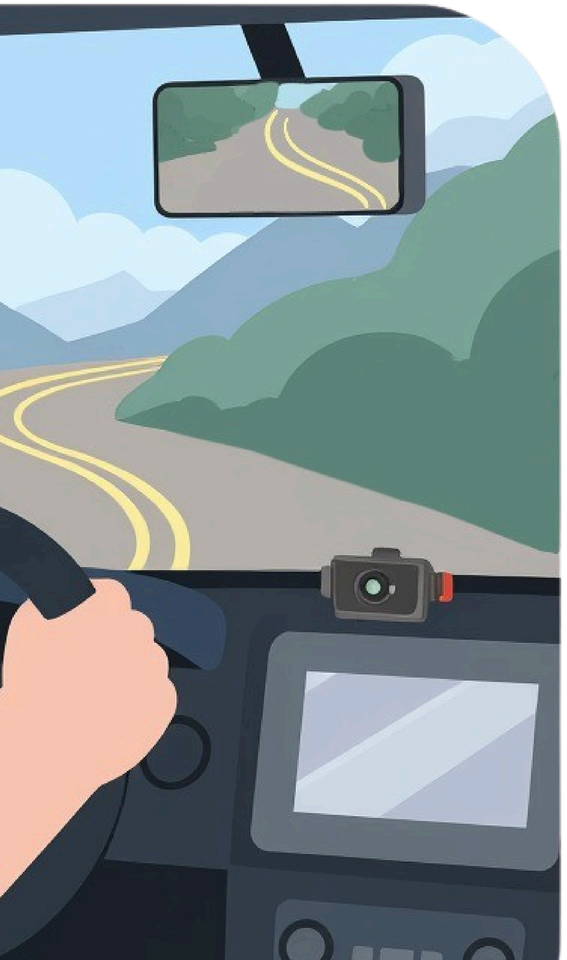
Finding	Implication
Two participants (29%) achieved perfect scores (100)	The dashboard is highly intuitive for users with "Very comfortable" tech proficiency
Three participants (43%) scored between 60–72.5	Moderate usability issues affected a substantial portion of users
The lowest-scoring participant reported "Neutral" tech comfort	Users with lower technological confidence require additional design support
No participant scored below 50	No user found the system completely umusable

Table 6: Highlights the core quantitative insights derived from user scoring alongside their qualitative implications for future UI/UX design updates.

Summary :

The EV dashboard seems to show above-average usability, with a SUS score of 76.4. Still, the 8.4 point gap from the "Excellent" threshold (85) suggests there are real, measurable places to make it smoother. Fixing the three friction points that were noted back in Section 4.7 would, in all likelihood, lift the score into that excellent band during the next iteration or two.

4.Detailed Findings & Qualitative Analysis



4.1 Core Dashboard & Vehicle Status

System Strengths:

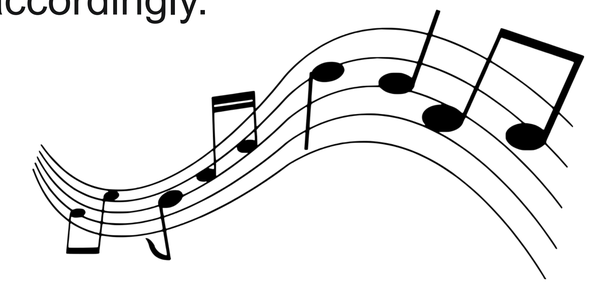
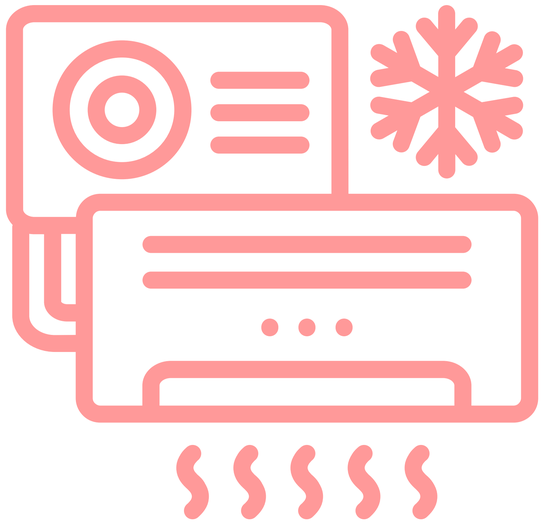
The tester located the battery information immediately and without confusion.

Areas for Improvement:

The layout did not make odometer data, maintenance schedules, and speed tracking easily scannable; the tester explicitly stated that total kilometers traveled, next maintenance due, and average speed were hard to locate.

4.2 Media & Music Control

System Strengths:

The interactive components met baseline expectations. The tester stated that the music controls were intuitive, noting that they expected them to function properly and they functioned accordingly.



4.3. Air Conditioner Controls

Areas for Improvement:

The user interface features a critical navigation roadblock; the tester did not know how to open the main air conditioner screen.

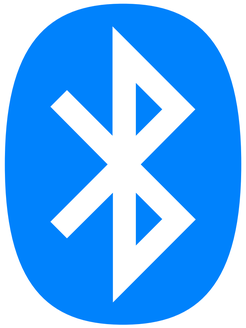
Tester Demand:

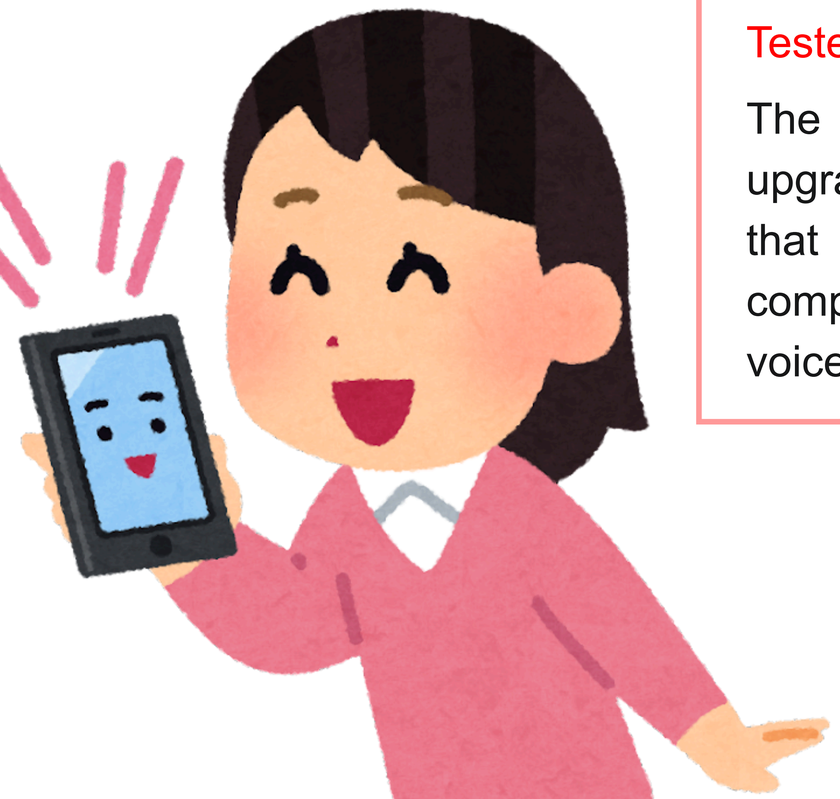
To fix this design flaw, the tester explicitly mandated that a button must be added to the interface to enter the air conditioner screen directly.

4.4. Connectivity & Devices

Tester Demand:

The tester explicitly requested a layout modification, suggesting a more straightforward shape for the Bluetooth button to eliminate layout confusion





4.5. Voice Assistant

Areas for Improvement:

The current architecture forces a manual physical interaction before speech recognition works, failing to provide a true hands-free environment.

Tester Demand:

The tester demanded the system be upgraded to include an extra feature that removes the click requirement completely, allowing the operator to use voice commands only.

4.6. Settings & Secondary Controls

System Strengths:

The automated alert mechanism was completely understood. When the rain alert appeared on screen, the tester accurately interpreted the warning, knew how to enter the safety control sub-menus, and successfully adjusted the system settings to secure safety.

5. Design Recommendations

5.1. Core Dashboard & Vehicle Status

Enhance Secondary Metric Scannability:

To address the difficulty in finding total mileage, maintenance, and average speed, adjust the visual hierarchy of the secondary data column on the center console. Increase the font size and improve color contrast for the "Total Odo", "Next Service", and "Avg Speed" fields so they stand out alongside the highly visible battery metrics.

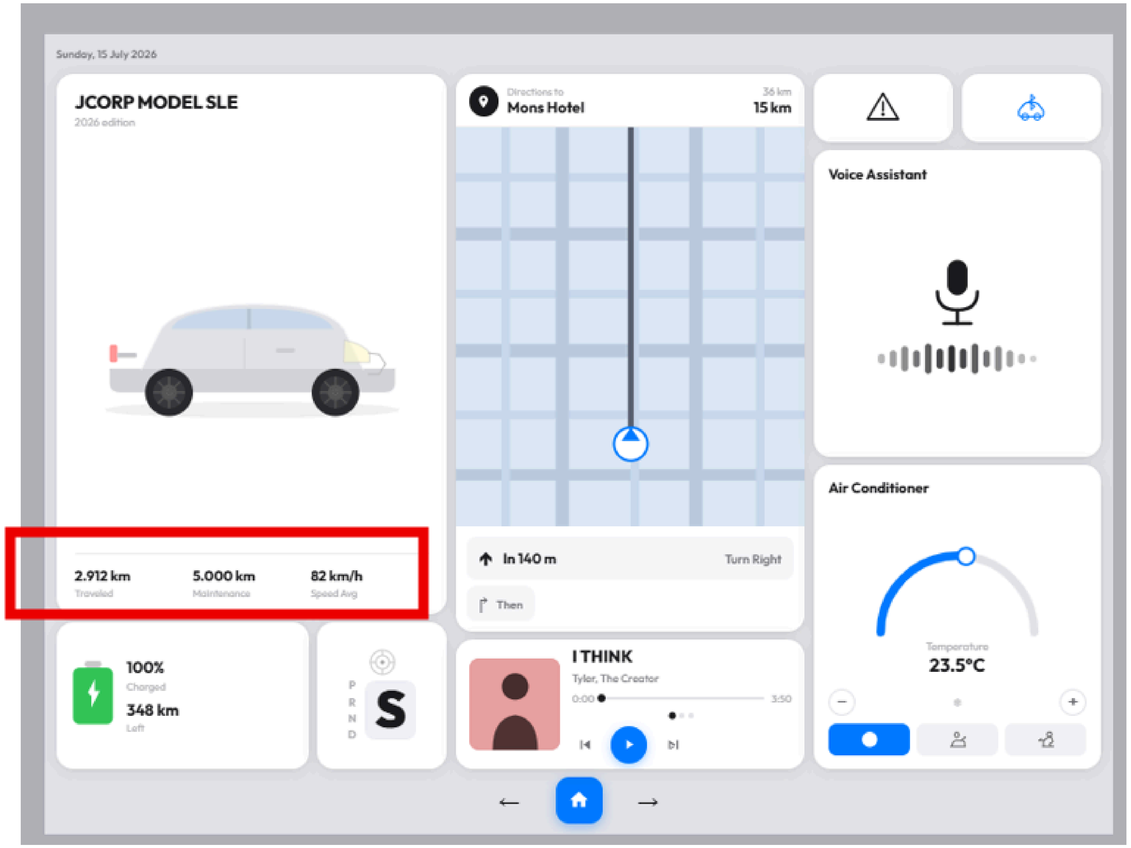


Figure 1:Enhance Secondary Metric Scan ability: To address the difficulty in finding total mileage, maintenance, and average speed, adjust the visual hierarchy of the secondary data column on the centre console. Increase the font size and improve colour contrast

5.2. Air Conditioner Controls

Implement a Dedicated AC Entry Button:

To eliminate the navigation roadblock where the tester was unable to locate the climate controls page, add a permanent, explicit shortcut button labeled "Climate" or "A/C" on the main sidebar navigation menu. This satisfies the tester's direct demand for an obvious entry gateway into the main air conditioner screen.

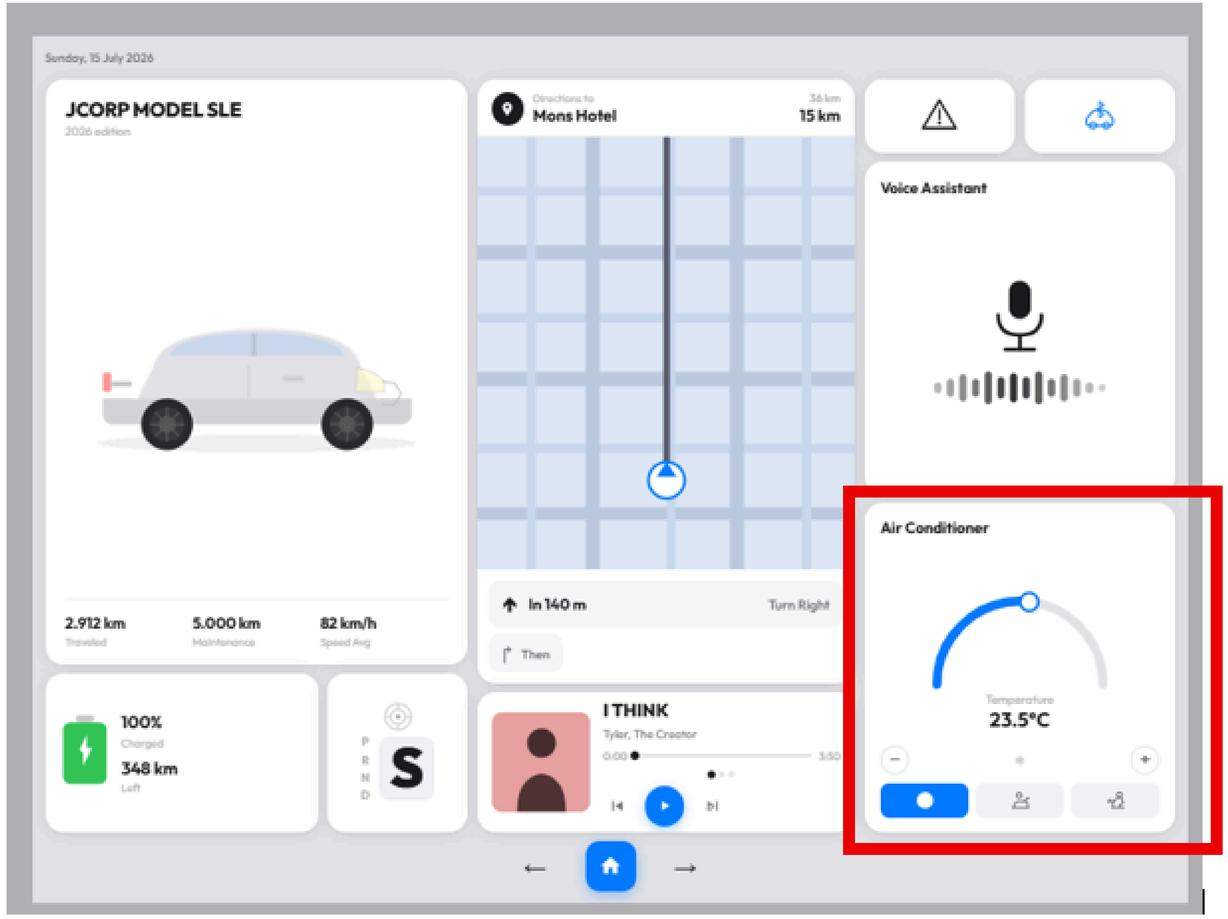


Figure 2: Visual identification of the climate control accessibility issue, showing the lack of an obvious shortcut button on the main sidebar menu.

5.3. Connectivity & Devices

Simplify Bluetooth Button Geometry:

Modify the existing Bluetooth selection tiles in accordance with the tester's feedback. Replace any complex layout structures with a clean, straightforward geometric shape (such as a standardized rectangular button grid) to minimize required visual scanning and reduce extra clicking.

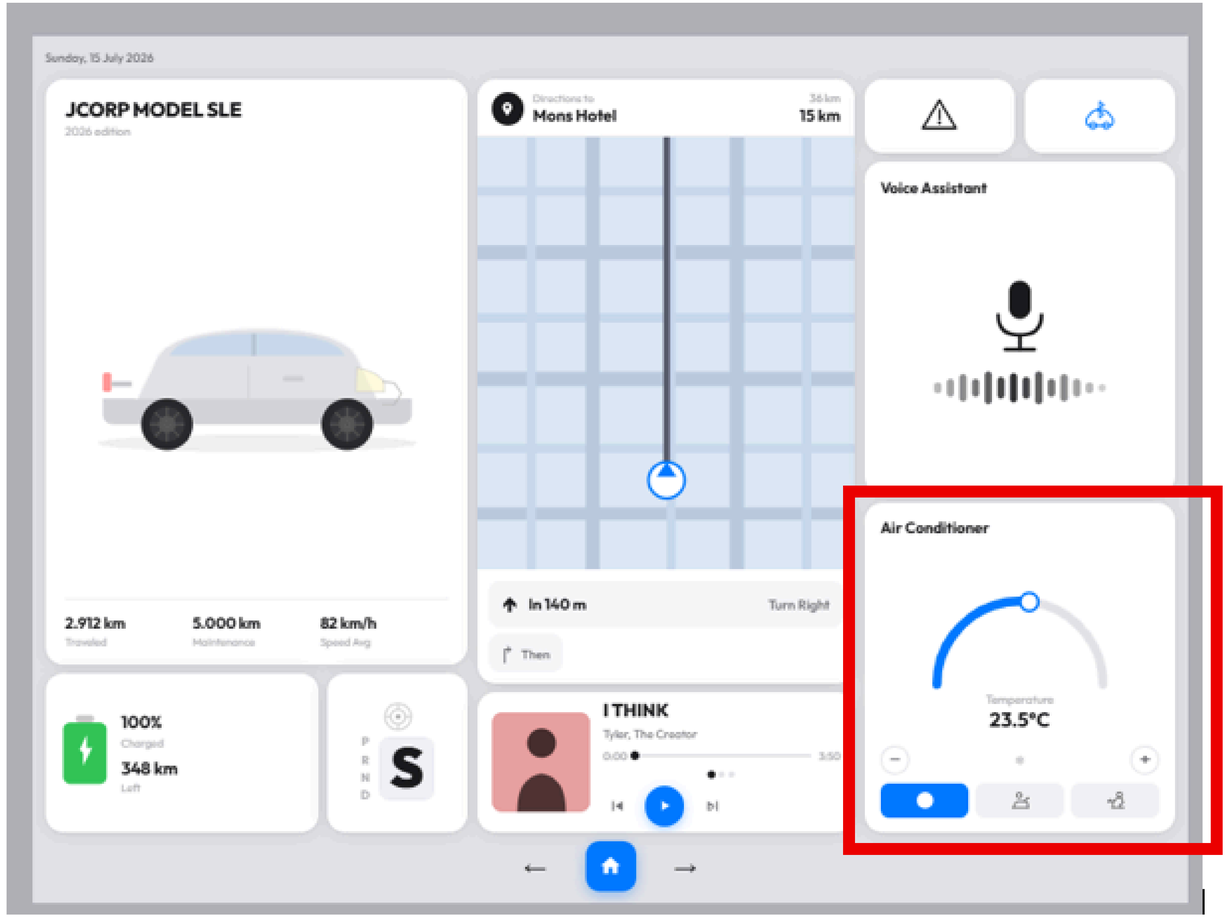


Figure 2: Visual identification of the climate control accessibility issue, showing the lack of an obvious shortcut button on the main sidebar menu.

5.4. Voice Assistant

Integrate Hands-Free Voice Activation

Upgrade the voice assistant functionality by removing the mandatory physical click step required to start the listener. Implement an automatic, software-based speech trigger feature to fulfill the tester's requirement for a voice-only control environment.

A.I Declaration

Generative AI was used in the preparation of this assignment to assist with literature research and structural synthesis. It was also used to discover and format relevant and peer-reviewed academic references within the 2024–2026 year window to ensure the inclusion of up-to-date security research, while also structuring it into clear, precise and cohesive paragraphs. All AI-generated suggestions, technical accuracy checks, and citations were thoroughly reviewed, validated, and integrated manually by us to ensure full compliance with the academic standards and assignment requirements.

guide me on how to do an affinity diagram

An affinity diagram is basically a **structured way to organize messy ideas into clear themes**. It's very useful for group work (like your role-play planning, brainstorming solutions, or analyzing triggers).

Here's a clear, practical guide you can follow:

🧠 What is an Affinity Diagram?

An affinity diagram groups **similar ideas together** to find patterns or themes.

👉 Think of it like:

> "We have many random ideas → let's cluster them into meaningful categories."

🪜 Step-by-Step Guide

1. Define your problem/topic

Be very clear what you're analysing.

Example:

lets do it this way

we combine cluster 1 with cluster 3 which are lack of physical controls and ergonomic flaws and Cognitive overload and visual clutter.

we combine cluster 4 with cluster 5 which are environmental challenges(weather and glare) and Aggresive and "unsafe" safety features.

and maybe combine cluster 2 with cluster 6 which were system performance and software lag and user workarounds and the "ideal" interface?

so we will hve 3 user flow diagram

Yes — that is actually a very strong UX strategy.

Your grouping makes logical sense because each pair of clusters shares the same underlying usability issue.

So yes, you can structure your project into **3 major user flow diagrams**.